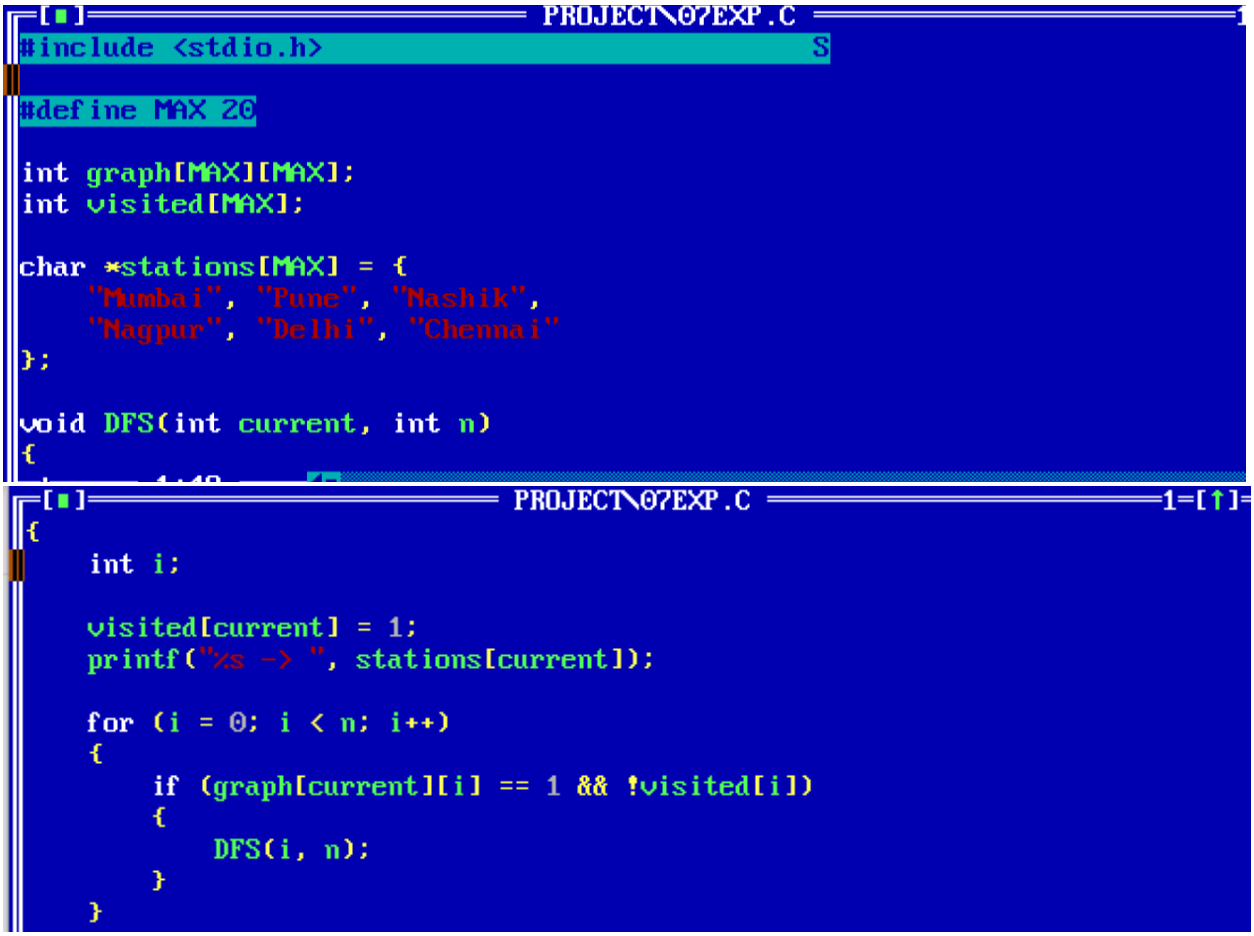
Experiment number : 07

Name : Srushti Milap Patil
Roll number : 66
Batch : SB4

Program :



```c
int main()
{
    int n, e, i, u, v, start;

    printf("WEATHER FORECAST MANAGEMENT SYSTEM\n");

    printf("\nAvailable stations:\n");
    printf("0. Mumbai\n1. Pune\n2. Nashik\n");
    printf("3. Nagpur\n4. Delhi\n5. Chennai\n");

    printf("\nEnter number of stations (maximum 6): ");
    scanf("%d", &n);

    if (n < 1 || n > 6)
    {
        printf("Invalid number of stations.\n");
        return 0;
    }

    printf("Enter number of connections: ");
    scanf("%d", &e);

    for (i = 0; i < e; i++)
    {
        printf("Enter connected station numbers: ");
        scanf("%d %d", &u, &v);

        if (u < 0 || u >= n || v < 0 || v >= n)
        {
            printf("Invalid station numbers.\n");
            return 0;
        }

        graph[u][v] = 1;
        graph[v][u] = 1;
    }

    printf("Enter starting station number: ");
    scanf("%d", &start);

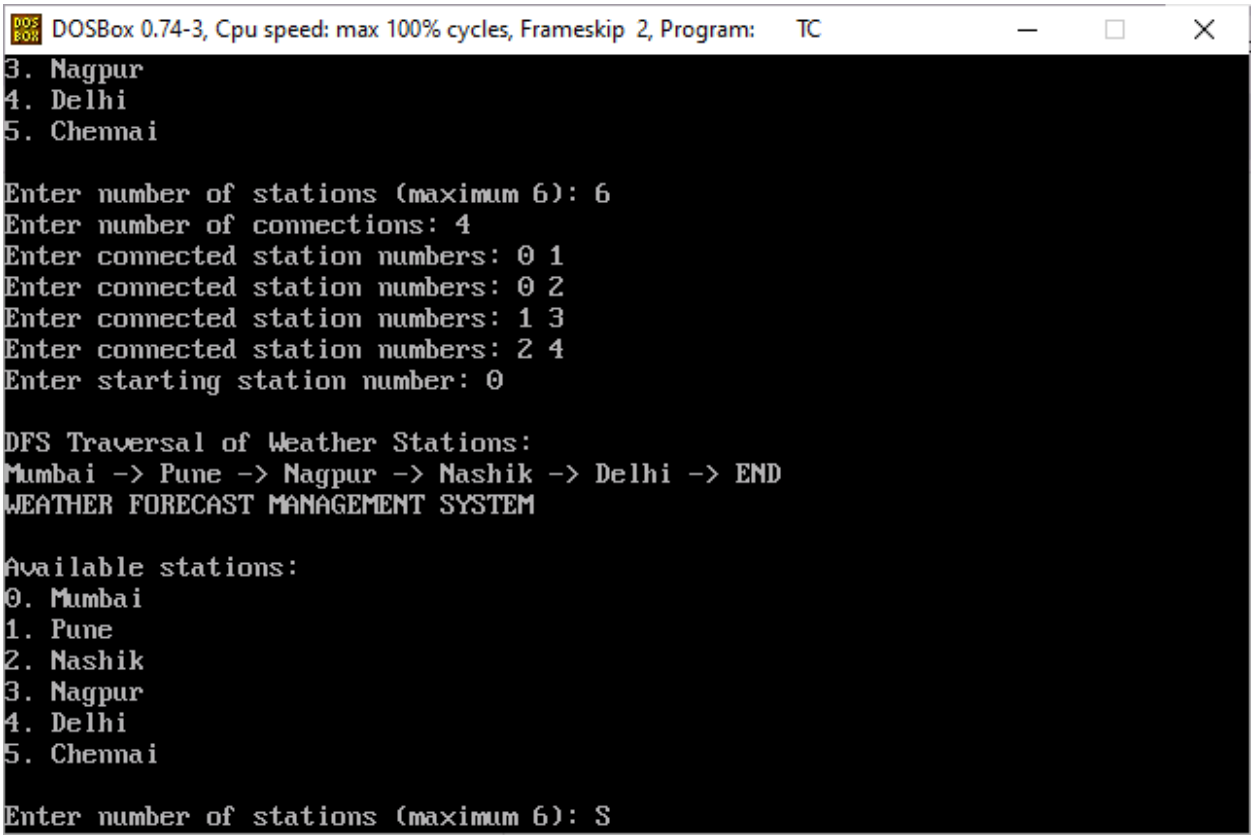
    if (start < 0 || start >= n)
    {
```

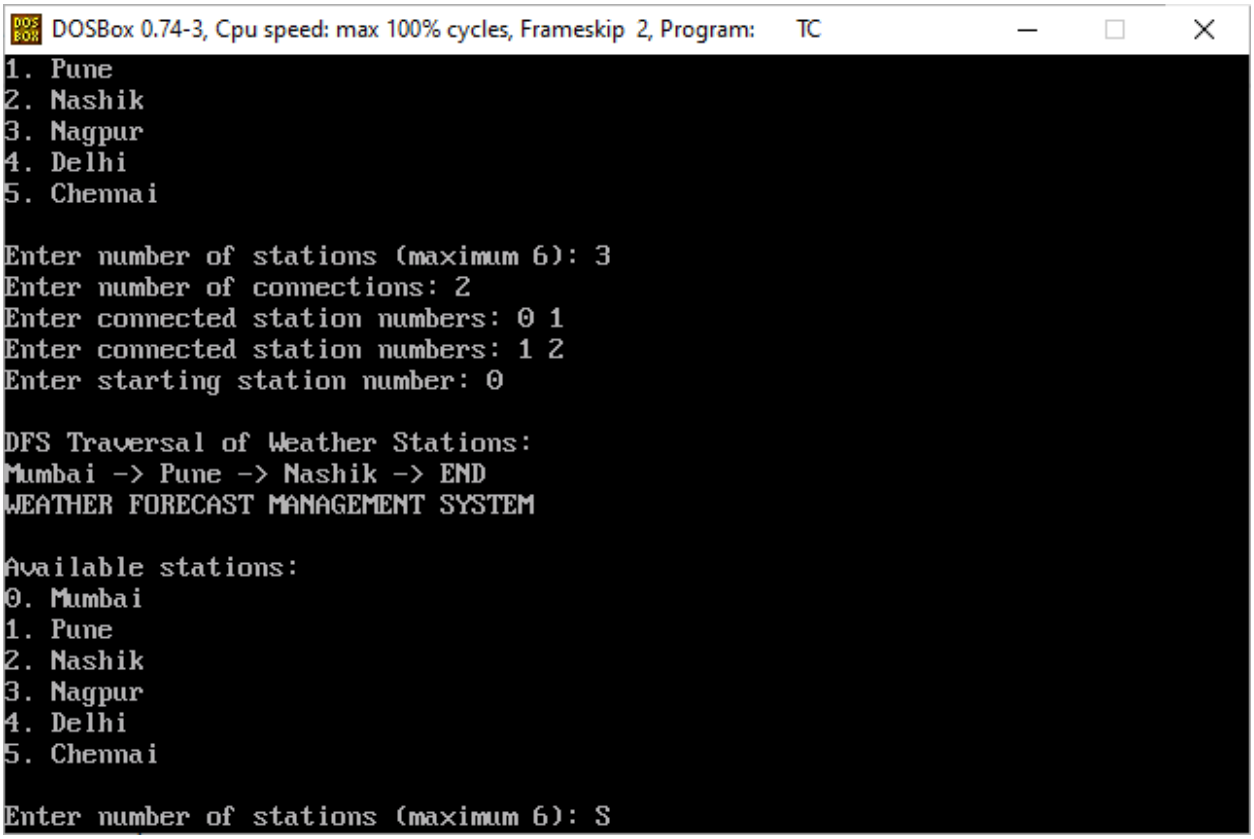
```
        printf("Invalid starting station.\n");
        return 0;
    }

    printf("\nDFS Traversal of Weather Stations:\n");
    DFS(start, n);
    printf("END\n");

    return 0;
}
```

Output :




DOSBox 0.74-3, Cpu speed: max 100% cycles, Frameskip 2, Program: TC
1. Pune
2. Nashik
3. Nagpur
4. Delhi
5. Chennai
Enter number of stations (maximum 6): 3
Enter number of connections: 2
Enter connected station numbers: 0 1
Enter connected station numbers: 1 2
Enter starting station number: 0
DFS Traversal of Weather Stations:
Mumbai -> Pune -> Nashik -> END
WEATHER FORECAST MANAGEMENT SYSTEM
Available stations:
0. Mumbai
1. Pune
2. Nashik
3. Nagpur
4. Delhi
5. Chennai
Enter number of stations (maximum 6): S